

WT EXPERIMENT - 4

Name	Class	Roll no.
Kaunal punjabi	D15A	43
Manav Punjabi	D15A	44
Hitesh Rohra	D15A	56

Aim: Component and Arduino Introduction

Introduction:

The Arduino Uno is a versatile and widely used microcontroller board that serves as the foundation for many electronics projects. It is an open-source platform that allows users to design and develop interactive electronic devices. The board consists of an Atmega328 microcontroller that can be programmed to control various hardware components, making it ideal for applications in automation, robotics, IoT, and more.

In this experiment, we will use an Arduino Uno to create a fire and smoke detection system. Fire and smoke detection systems play a crucial role in early fire warning and prevention. This experiment focuses on building a low-cost, real-time fire and smoke detection system using an Arduino Uno. The system utilizes a fire sensor to detect flames and a smoke sensor to measure air quality. If dangerous levels of fire or smoke are detected, the LCD display alerts the user, and a buzzer is triggered. This project can be implemented in residential areas, offices, and industrial setups to prevent fire-related hazards.

Arduino Uno Overview

The Arduino Uno is one of the most popular boards in the Arduino family due to its simplicity, low cost, and ease of use. It features:

- **Microcontroller:** Atmega328P
- **Operating Voltage:** 5V
- **Digital I/O Pins:** 14 (6 can be used as PWM outputs)
- **Analog Input Pins:** 6
- **Flash Memory:** 32 KB
- **Clock Speed:** 16 MHz
- **USB Connection:** For programming and communication
- **Power Supply:** Can be powered via USB or an external power supply

The Arduino Uno can be programmed using the Arduino IDE, which allows users to write and upload code that controls various electronic components connected to the board.

Installation and Setup

Installing the Arduino IDE

1. Download and install the Arduino IDE from [Arduino's official website](#).
2. Connect the Arduino Uno to your computer via USB.
3. Open the Arduino IDE and select the correct board under **Tools > Board > Arduino Uno**.
4. Select the correct port under **Tools > Port**.
5. Upload a sample program to verify the setup.

Running a Test Blink Program

```
void setup() {  
  
  pinMode(LED_BUILTIN, OUTPUT);  
  
}  
  
void loop() {  
  
  digitalWrite(LED_BUILTIN, HIGH);  
  
  delay(1000);  
  
  digitalWrite(LED_BUILTIN, LOW);  
  
  delay(1000);  
  
}
```

If the built-in LED blinks, your setup is working fine.

Components Used

1. Arduino Uno – Main microcontroller board.
2. Fire Sensor (Flame Sensor Module) – Detects fire using infrared waves.
3. Smoke Sensor (MQ-2 or MQ-135) – Measures smoke concentration in the air.
4. Buzzer – Provides an audio alarm in case of fire or smoke detection.
5. LCD Display (16x2, I2C) – Shows real-time fire and smoke levels.
6. Jumper Wires – For connecting the components.
7. Breadboard – For easy circuit prototyping.

Code:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// LCD I2C Address (Agar 0x27 kaam na kare toh 0x3F try karo)
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Pin Definitions
const int fireSensor = 8; // Fire sensor D0 → Pin 8
const int smokeSensor = A0; // Smoke sensor A0 → A0
const int buzzer = 9; // Buzzer → Pin 9

void setup() {
  pinMode(fireSensor, INPUT);
  pinMode(smokeSensor, INPUT);
  pinMode(buzzer, OUTPUT);

  lcd.init(); // Initialize LCD
  lcd.backlight(); // Turn on LCD backlight

  Serial.begin(9600);
}

void loop() {
  int fireValue = digitalRead(fireSensor); // Fire Sensor D0 output (0 or 1)
  int smokeValue = analogRead(smokeSensor); // Smoke Sensor Analog Value

  // Serial Monitor Debugging
  Serial.print("Fire: ");
  Serial.println(fireValue);
  Serial.print("Smoke: ");
  Serial.println(smokeValue);

  lcd.clear();
  lcd.setCursor(0, 0);

  // Fire Detection with Levels
  if (fireValue == 0) { // Fire Detected (LOW means fire present)
    lcd.print(" FIRE: HIGH!!!");
    digitalWrite(buzzer, HIGH); // Buzzer ON
  } else {
    lcd.print("Fire: Normal");
    digitalWrite(buzzer, LOW); // Buzzer OFF
  }

  // 🌫️ Smoke Detection with Levels
  lcd.setCursor(0, 1);
  if (smokeValue > 55) { // HIGH Smoke Level
    lcd.print("Smoke: HIGH ");
    digitalWrite(buzzer, HIGH); // Buzzer ON
  } else if (smokeValue > 45) { // MEDIUM Smoke Level
    lcd.print("Smoke: MEDIUM");
    digitalWrite(buzzer, LOW); // Buzzer OFF
  } else if (smokeValue > 25) { // LOW Smoke Level
```

```
    lcd.print("Smoke: LOW");  
    digitalWrite(buzzer, LOW); // Buzzer OFF  
  } else {  
    lcd.print("Smoke: Normal");  
    digitalWrite(buzzer, LOW);  
  }  
  
  delay(1000); // Wait for 1 sec before next reading  
}
```

Code Breakdown

Libraries

- `#include <Wire.h>`
- `#include <LiquidCrystal_I2C.h>`

Pin Setup

- **Fire Sensor:** Digital Pin **8**
- **Smoke Sensor:** Analog Pin **A0**
- **Buzzer:** Digital Pin **9**

Setup Function

- Initializes LCD, sets up pin modes, and begins serial communication.

Loop Function

1. Reads **fire sensor** (digital value: **0** = fire detected, **1** = no fire).
2. Reads **smoke sensor** (analog value: higher values = more smoke).
3. Displays **fire & smoke levels** on the **LCD screen**.
4. Activates **buzzer** when **fire or high smoke levels** are detected.

Methodology:

Fire Detection

- The fire sensor detects fire using infrared (IR) radiation.
- If a fire is detected, the LCD shows a warning, and the buzzer rings.

Smoke Detection

- The smoke sensor (MQ-2 or MQ-135) measures air quality.
- If high smoke levels are detected, a warning is displayed, and the buzzer rings.
- Moderate smoke levels trigger a notification, but the buzzer remains off.

LCD & Buzzer Alerts

- LCD Display provides real-time fire and smoke levels.
- Buzzer sounds when a fire or high smoke concentration is detected.

Conclusion

The Fire and Smoke Detection System successfully detects fire and various levels of smoke using an Arduino Uno, a fire sensor, and a smoke sensor. The system provides real-time monitoring through an LCD display and alerts users with a buzzer when fire or high smoke concentration is detected.